TokenOptimizer

Architectural Design Document

Token-optimisation platform for AI-assisted development — layered, offline-first, and measurable.

Version 0.1.0 • 15 Jul 2026 • Status: Baseline

Document Control

Field	Value
Title	TokenOptimizer — Architectural Design Document
Version	0.1.0
Date	15 Jul 2026
Status	Baseline
Audience	Engineers, architects, and reviewers
Scope	Full system: optimizer engine, feature-dev skills, interfaces, evaluation
Related	docs/FEATURES.md, docs/CHANGELOG.md, README.md

Contents

1. Introduction & Purpose

TokenOptimizer reduces the number of tokens sent to a Large Language Model, cutting cost and latency per call while preserving every fact a downstream agent needs. Beyond a general text optimizer, it ships two **feature-development skills** — PRD compression and codebase anchoring + model routing — with an offline A/B suite that proves the cost/quality outcome.

This document describes the system's architecture: its layers, components, runtime flows, cross-cutting concerns, and the design decisions behind them. The guiding principle is **offline-first**: the deterministic strategies and both skills run with no API key; a Claude key or a local model only improves quality, and is never required.

The architecture and flow figures in this document are generated from the PlantUML sources under `docs/diagrams/*.puml` (scalable SVG versions live alongside the PNGs used here), so they stay in lock-step with the code they describe.

2. Architectural Goals & Constraints

Goals

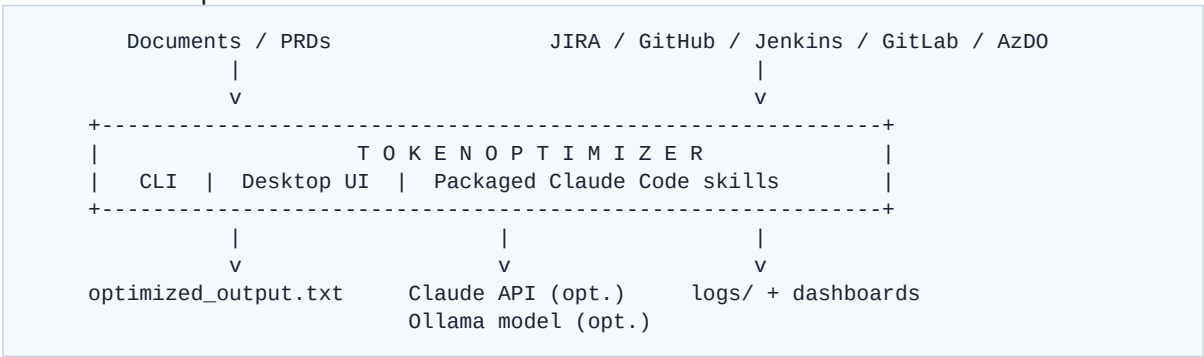
- Reduce LLM token spend measurably (input compression, routing, caching).
- Preserve decision-critical content — never silently drop facts or acceptance criteria.
- Work fully offline; degrade gracefully when optional dependencies are absent.
- Ground AI output in verifiable code (file:line) to eliminate hallucination rework.
- Be observable and reproducible: per-run logs and deterministic evaluation.

Constraints

- Python 3.10+; minimal hard dependencies (anthropic, httpx, python-dotenv, python-docx).
- Desktop UI must be dependency-free (standard-library Tkinter only).
- No secrets in logs or output; credentials only via environment / .env.
- Optional extras: tiktoken (exact offline counts), reportlab (docs), Ollama (local model).

3. System Context

TokenOptimizer sits between a user (or automation) and external systems. Inputs are documents, JIRA issues, GitHub PRs, or PRDs; outputs are optimized text, anchored plans, routing decisions, run logs, and validation reports.



4. Logical Architecture (Layered View)

The package is organised into layers. Higher layers depend on lower ones; the `core` layer is depended upon by everything and depends on nothing else in the project.

```
+-------------------------------------------------------------+
|  INTERFACES     cli.py   .   ui/  (Tkinter desktop app)     |
+-------------------------------------------------------------+
|  SKILLS (top-level plugins)   prd_compression .             |
|                 codebase_anchoring . model_routing          |
|  WORKFLOWS      automations/ (triage, review)               |
|  EVALUATION     evaluation/ (ab_runner, rubric, cost, dashboard)|
+-------------------------------------------------------------+
|  OPTIMIZE       local . compress . summarize . prefilter .  |
|                 tokens . text/structured pipelines          |
+-------------------------------------------------------------+
|  INTEGRATIONS   jira . github . gitlab . jenkins . azdo . docs  |
+-------------------------------------------------------------+
|  CORE           config  .  run_log  .  llm (client + cache) |
+-------------------------------------------------------------+
```

- **Core** — cross-cutting infrastructure: configuration, per-run logging, Claude client + response cache.
- **Integrations** — turn external systems and files into plain text.
- **Optimize** — the reduction strategies and the pipelines that compose them.
- **Skills / Workflows / Evaluation** — feature-dev skills, ready-made automations, and the A/B harness.
- **Interfaces** — CLI, desktop UI, and packaged Claude Code skills.

Figure 1 expands this into a component view: how the interfaces resolve configuration, fan out to the source connectors, feed the ordered optimize strategies, and converge on the Claude client and its two caching layers before results, logs, and dashboards are emitted.

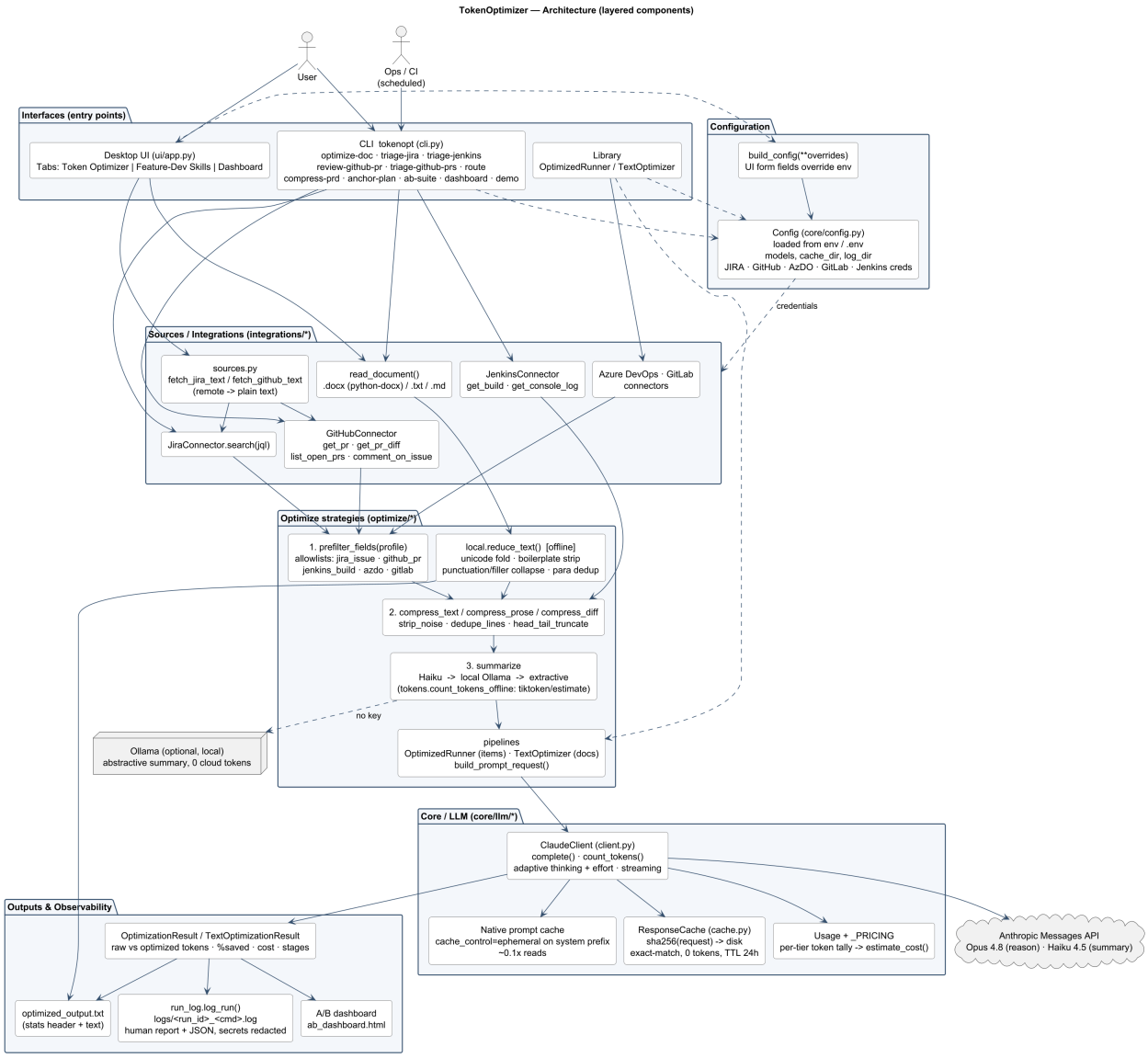


Figure 1 — Layered component architecture (interfaces -> config -> sources -> optimize -> core/LLM -> outputs).

5. Component Responsibilities

Package	Responsibility
core/config.py	Env-driven configuration (models, keys, cache/log dirs).
core/run_log.py	Write one structured, secret-free record per run.
core/llm/	Claude client (count_tokens, adaptive thinking) + disk response cache.
optimize/	Deterministic reductions, compression, summarization tiers, token counting, and the text/structured pipelines.
integrations/	Connectors for JIRA, GitHub, GitLab, Jenkins, Azure DevOps, and documents.
automations/	End-to-end flows: JIRA triage, Jenkins RCA, GitHub PR review/triage.
skills/prd_compression/	Skill 1 — compress a PRD into requirement atoms (top-level plugin).
skills/codebase_anchoring/	Skill 2a — index a repo and anchor plan steps to file:line.
skills/model_routing/	Skill 2b — classify task complexity and route to a model.

Package	Responsibility
evaluation/	A/B runner, 25-point rubric, cost model, dataset, HTML dashboard.
ui/	Tkinter desktop app (three tabs) — app.py, python -m entry.
cli.py	The tokenopt command exposing every capability.

5.1 Optimize strategies in detail

The four stacked strategies run in a fixed order so each stage feeds clean input to the next. Structured payloads (JIRA/PR/build items) enter through pre-filtering; free-form documents enter through the deterministic reductions.

Strategy / function	Detail
1. prefilter_fields(profile)	Field allowlists per profile drop everything irrelevant before serialization (0 tokens). Profiles: jira_issue, github_pr, jenkins_build, azdo_workitem, gitlab_issue. Nested user/status objects are collapsed to their display value.
2a. compress_text	Log/JSON path: strip_noise (INFO/DEBUG, deep stack frames, ANSI, timestamps; error/warn lines always kept) -> dedupe_lines (repeat counts) -> head_tail_truncate.
2b. compress_prose	Document path: collapse whitespace/blank runs, dedupe verbatim lines, truncate. Runs before summarization so duplicates can't dominate the summary.
2c. compress_diff	Review path: keep file/hunk headers and +/- lines; collapse unchanged context to '... N unchanged lines ...' markers. Largest single saving on a PR review.
3. summary tier	Best-available first: Claude Haiku (API key) -> local Ollama model (no cloud tokens) -> entity-anchored extractive (never drops error codes, versions, IDs, paths).
tokens.count_tokens	API count_tokens when a key is set; else tiktoken cl100k; else a char/word heuristic. The chosen method is recorded in the result and the run log.

6. Feature-Development Skills

Skill 1 — PRD Compression (skills/prd_compression)

Segments a PRD into units, classifies each into a requirement category (goal, acceptance-criteria, constraint, non-functional, dependency, out-of-scope), drops low-signal framing, and re-renders terse atoms. Acceptance criteria are preserved verbatim. ~67–73% input-token reduction.

Skill 2a — Codebase Anchoring (skills/codebase_anchoring)

Indexes a repository into a symbol table with exact file:line locations (AST for Python, regex for JS/TS/Java/Go/Ruby/C#) and resolves each plan step's symbol mentions to a real anchor. Unresolved symbol-like terms are flagged as possible hallucinations.

Skill 2b — Model Routing (skills/model_routing)

Classifies a task as trivial / standard / complex with a confidence score, then routes to haiku / sonnet / opus respectively. A confidence-threshold fallback upgrades one tier toward premium when the

classifier is unsure, so a cheap model is never chosen on a coin-flip.

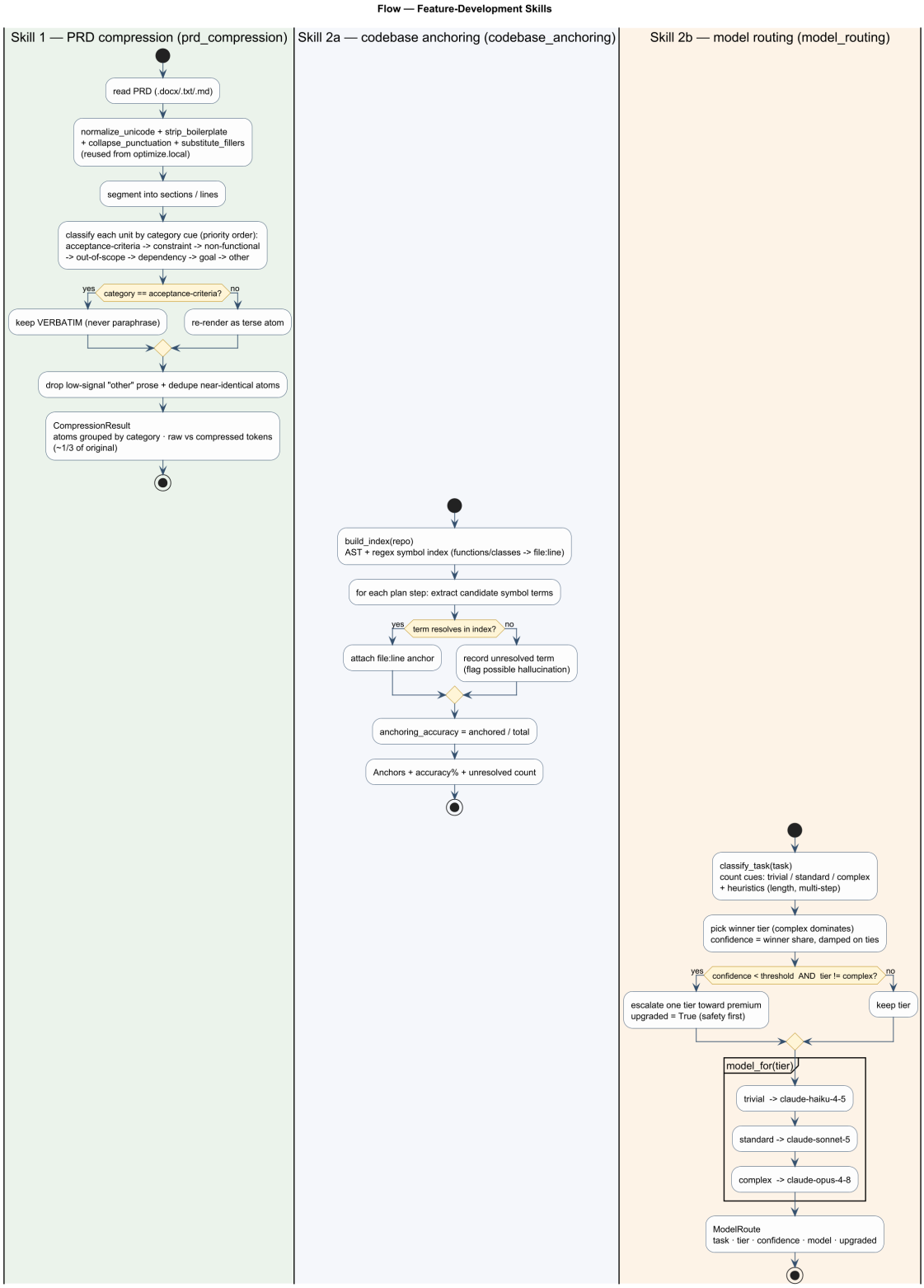


Figure 2 — Feature-development skill flows: PRD compression (category-priority, acceptance criteria verbatim), codebase anchoring (file:line resolve, hallucination flags), and confidence-based model routing.

7. Key Runtime Flows

Every command shares one spine — acquire, optimize, measure, cache-or-call, report — and specializes only the acquire and prompt steps. The following figures trace that spine and the three concrete pipelines built on it.

7.1 End-to-end data flow (the shared spine)

The interface resolves a `Config` (env / .env, with UI form overrides), acquires text from a document or a connector, runs the ordered optimize strategies, counts raw and optimized tokens, then either returns a local-cache hit (0 tokens) or calls the API with the stable system prefix marked for native prompt caching. Results, logs, and optional artifacts (PR comment, dashboard) are emitted last.

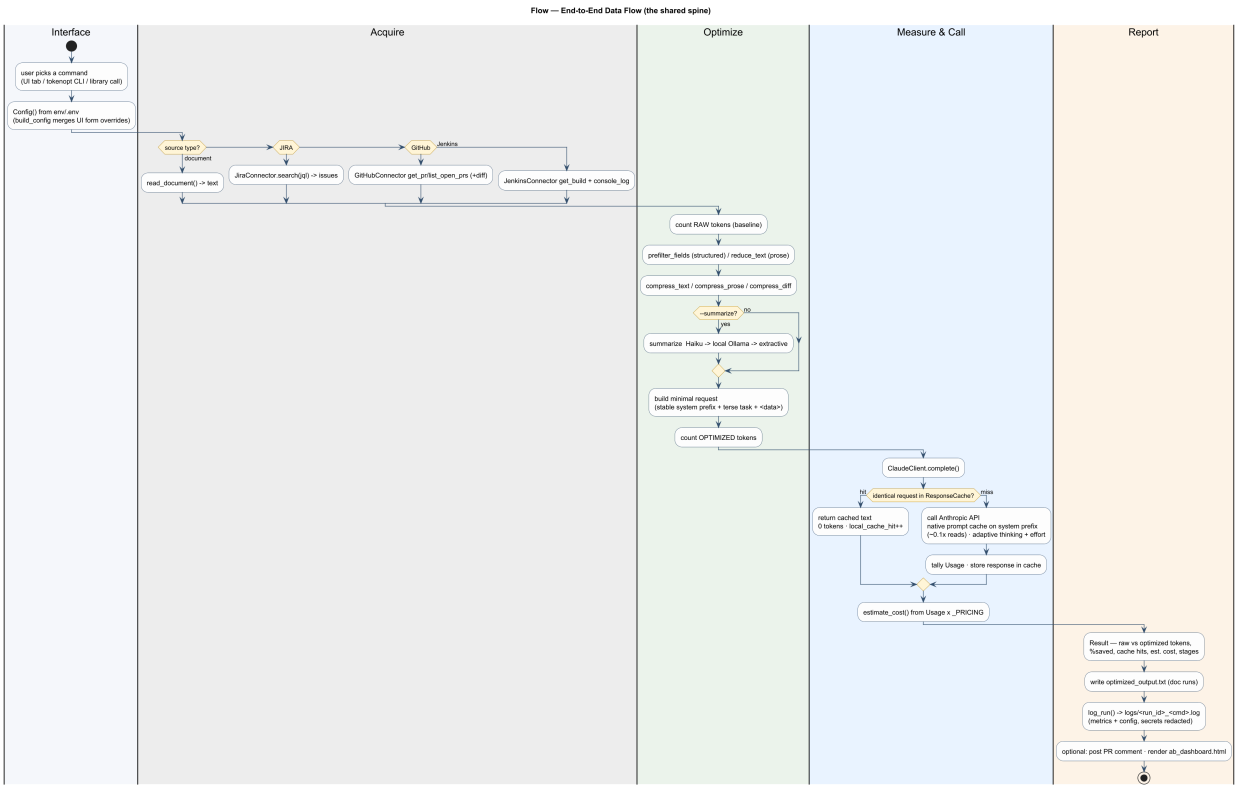


Figure 3 — The shared spine every command runs through, shown as swimlanes over the owning layer.

7.2 Document optimization (TextOptimizer.optimize)

Free-form document text runs the deterministic reductions (unicode fold, boilerplate strip, punctuation/filler collapse, paragraph dedup), then a whitespace/dedupe compression pass, then an optional summary whose tier is chosen by what is available. Token counting falls back API -> tiktoken -> heuristic, and the chosen method, stages, and summary tier are all reported.

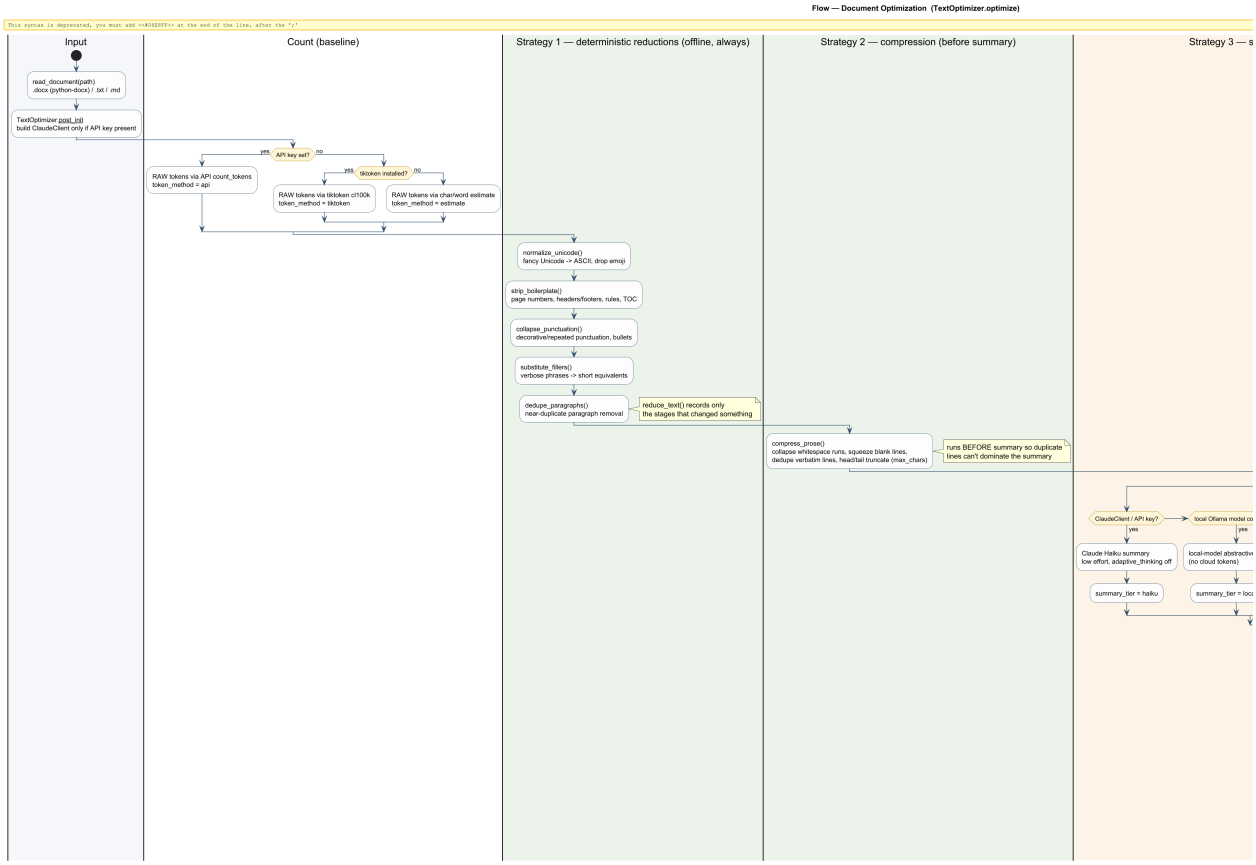


Figure 4 — Document optimization flow, including the token-method and summary-tier decision points.

7.3 Structured runner + caching (OptimizedRunner.run)

Structured items are pre-filtered by profile, compressed, and optionally summarized, then joined into a single user turn. The system prefix is sent as an ephemeral cacheable block; an exact-match on-disk cache short-circuits identical requests for 0 tokens, and usage is tallied per cache tier to estimate cost against per-model list prices.

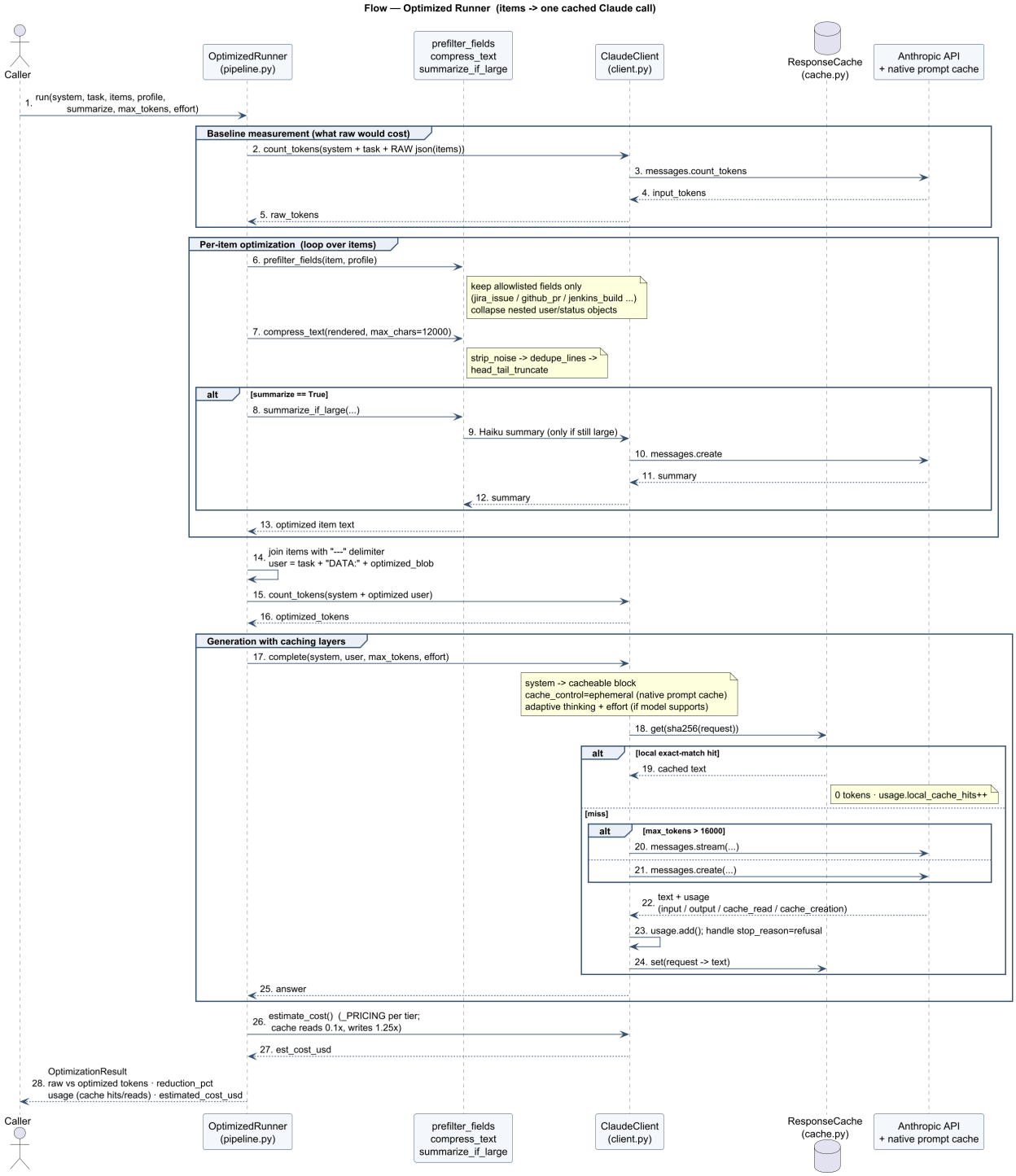


Figure 5 — Structured item pipeline and the two caching layers (local exact-match response cache + native prompt cache).

7.4 Automations (JIRA / Jenkins / GitHub)

Each automation acquires from its connector, applies the compression best suited to its payload (aggressive log compression for Jenkins, diff-aware compression for GitHub), then defers to the shared runner. The GitHub reviewer can post its review back to the PR thread; every run is logged.

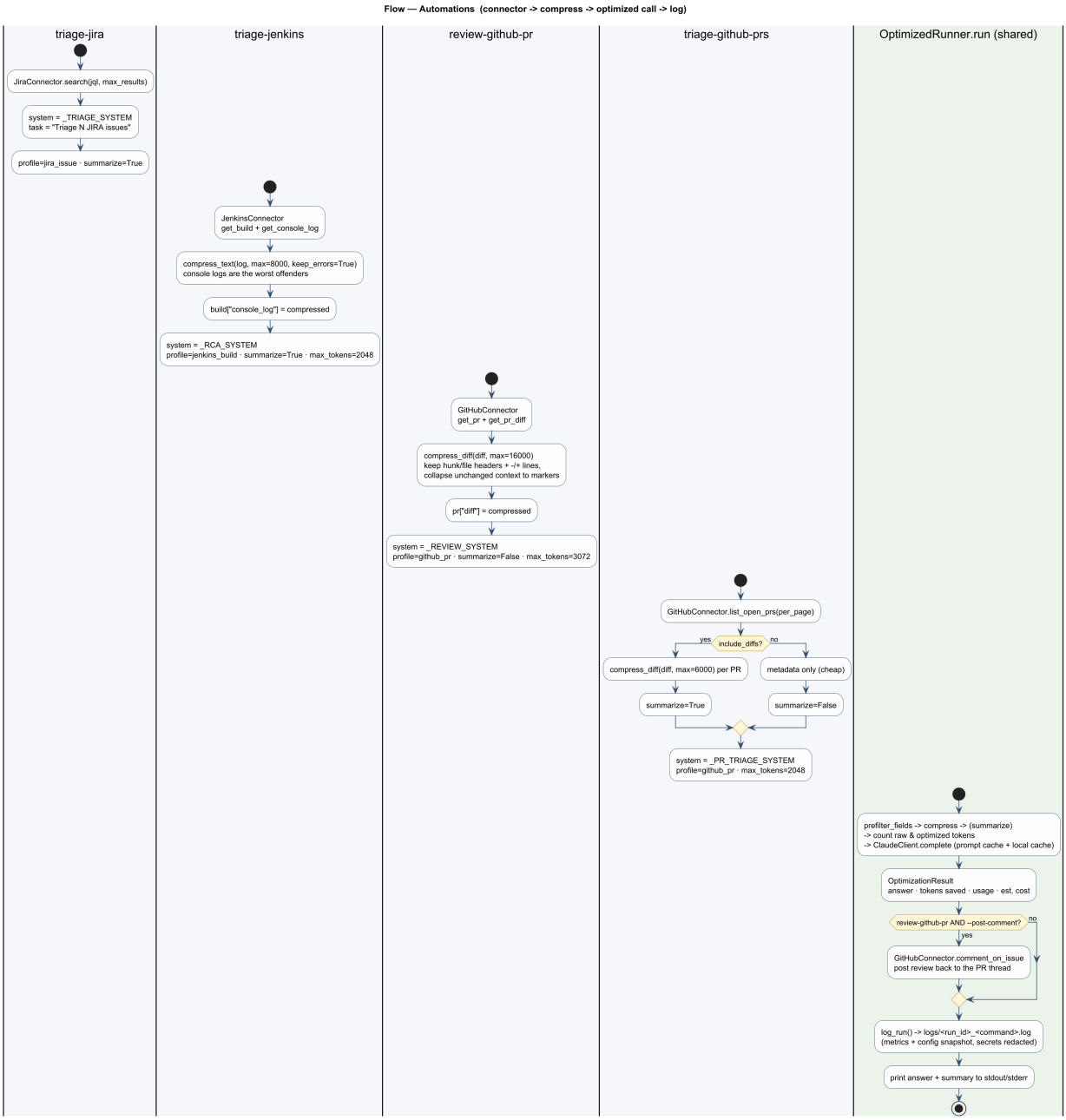


Figure 6 — Automation flows converging on the shared OptimizedRunner, with per-command system prompts, profiles, and settings.

8. Cross-Cutting Concerns

Concern	Approach
Configuration	core/config.py reads env / .env; blanks fall back to defaults. build_config() lets UI form fields override env per run.
Logging	core/run_log.py writes a per-run report + JSON block; never raises (returns '' on failure); API keys/tokens recorded only as booleans.
Caching (local)	core/llm/cache.py hashes the full request (sha256, sorted keys) to an on-disk JSON entry with a 24h TTL; identical calls return for 0 tokens.

Concern	Approach
Caching (native)	The stable system prefix is sent as a cache_control=ephemeral block so Claude bills reused prefixes at ~0.1x; volatile content stays in the user turn.
Cost accounting	Usage tallies input/output/cache-read/cache-creation tokens; estimate_cost applies per-tier list prices (reads 0.1x, writes 1.25x).
Token counting	optimize/tokens.py: API count_tokens -> tiktoken cl100k -> heuristic.
Generation control	Adaptive thinking + effort (only for models that accept it); streaming when max_tokens is large; stop_reason=refusal handled explicitly.
Error handling	Graceful degradation at every optional boundary; nothing hard-fails offline.
Security	No secrets in logs or output; credentials only via env / .env.

9. Evaluation & Validation Architecture

The evaluation package proves the cost/quality claims deterministically and offline. For each of 8 PRDs across 2 business units it runs a baseline arm (raw PRD -> premium model, no anchoring) against an optimised arm (compressed -> anchored -> routed) and scores both on a 25-point rubric.

- ab_runner.py — orchestrates both arms and aggregates deltas.
- quality_rubric.py — correctness, completeness, anchoring, actionability, no-hallucination.
- cost.py — tokens x per-model list price.
- datasets.py — 8 realistic verbose PRDs across Payments and Platform BUs.
- dashboard.py — self-contained HTML (inline SVG) rendered from the results.

Validated, test-locked outcome: ~73% average PRD compression, ~58% average cost savings, 24.0/25 average optimised quality (baseline 19.5).

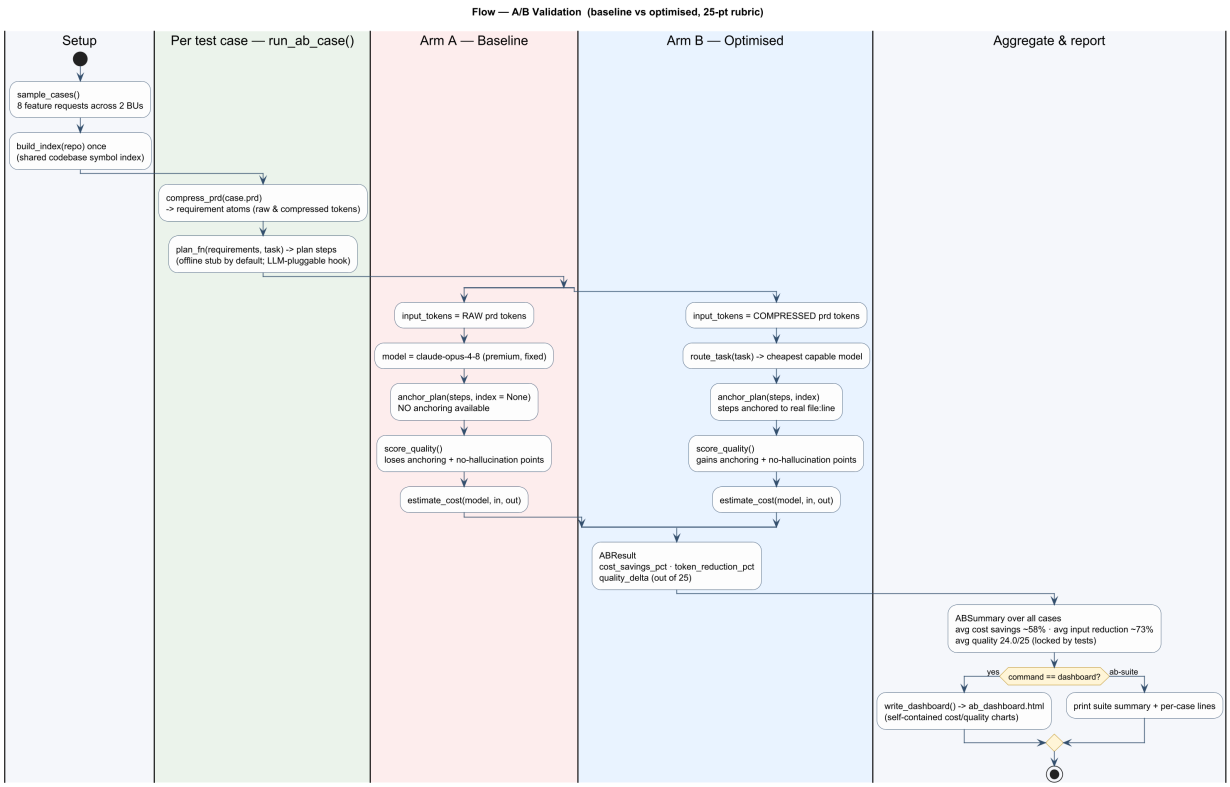


Figure 7 — A/B validation flow: baseline (raw PRD, premium model, no anchoring) vs optimised (compressed, routed, anchored), scored on the 25-point rubric and aggregated into the dashboard.

10. Deployment & Runtime View

- Installed as an editable package; the tokenopt console script is the primary entry point.
- Desktop UI: tokenopt ui or python -m token_optimizer.ui (Tkinter, no extra deps).
- Windows launcher run.bat wraps the venv for double-click or terminal use.
- Each skill is a self-contained top-level package under skills/ (code + SKILL.md + tests).
- Runs on a single machine; no server component. Optional network calls: Claude API, Ollama, integrations.

11. Technology Stack

Area	Choice
Language / runtime	Python 3.10+
Packaging	setuptools src-layout; console_scripts entry point
LLM SDK	anthropic (Claude); Messages API + prompt caching
HTTP	httpx (integration connectors)
Documents	python-docx
Token counting	tiktoken (optional), API count_tokens, heuristic fallback
Desktop UI	Tkinter (standard library)
Docs / PDF	reportlab (optional extra)
Local model	Ollama HTTP server (optional)
Testing	pytest (offline, no API key)

12. Key Design Decisions (Rationale)

Decision	Rationale
Offline-first pipeline	Real token savings without a key; API/local model only improves quality.
core / skills separation	Isolate cross-cutting infrastructure from product skills; clear dependency direction.
Deterministic skills & rubric	Reproducible results and testable claims; no model-in-the-loop nondeterminism.
Confidence-threshold routing	Under-powering a hard task costs more than the routing saved; upgrade when unsure.
Anchoring flags, not silently accepts	Surfacing unresolved references eliminates hallucination-driven rework.
src-layout package	Prevents accidental imports of the working tree; standard for distributable packages.

Decision	Rationale
Self-contained HTML dashboard	Zero runtime chart dependency; portable evidence for any stakeholder.
Skip-if-noop stages	Reported stages reflect what actually changed the text; easier to reason about.

13. Quality Attributes

- **Reliability** — graceful degradation at every optional boundary; logging can't break a run.
- **Security** — secrets never logged or emitted; credentials only via env / .env.
- **Testability** — 58 offline tests; headline claims locked by dedicated tests.
- **Performance / cost** — pre-filtering and compression are 0-token; routing and caching cut spend.
- **Maintainability** — layered packages, single-responsibility modules, relative imports.
- **Observability** — structured per-run logs (human + JSON) and an A/B dashboard.

14. Directory Structure

```
src/token_optimizer/     core package
  cli.py                 tokenopt command (all subcommands)
  core/                  config . run_log . llm (client + cache)
  optimize/              local . compress . summarize . prefilter .
                         tokens . text/structured pipelines
  integrations/          jira . github . gitlab . jenkins . azdo . docs
  automations/           triage . github review
  evaluation/            ab_runner . quality_rubric . cost . datasets .
                         dashboard
  ui/                    app.py (three-tab Tkinter app), __main__.py
skills/                  self-contained skill plugins (code + SKILL.md + tests)
  prd_compression/       Skill 1 . codebase_anchoring/  Skill 2a
  model_routing/         Skill 2b
docs/                    FEATURES.md . CHANGELOG.md . this ADD . plan . PDFs
examples/                sample PRD / document inputs
tests/                   test_optimize.py . test_hackcelerate.py
```

15. Risks & Future Work

- PRD compression is tuned for PRD-shaped input; raw Jira bug tickets suit the structured pipeline better.
- The A/B plan generator is a deterministic stub; a live LLM plan_fn can be plugged in without harness changes.
- Regex symbol indexing (non-Python languages) is heuristic; a tree-sitter backend could improve accuracy.
- Cost figures use list prices; wire real usage metering for production dashboards.
- Consider persisting dashboards and trend history across runs for longitudinal reporting.